ALEJANDRO GARCIA BALTAZAR

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OBJECTIVE

CS graduate actively transitioning into firmware and embedded systems. Hands-on experience with baremetal STM32 and ESP32 development, I2C peripheral integration, ADC sensor reading, MOSFET switching, and ESP-IDF. Background in technical support and full-stack development. Open to junior firmware and embedded roles.

EDUCATION

University of Arkansas, Grantham **August 2024**
Bachelor's of Science, Computer Science
Relevant Coursework:

- Data Structures and Algorithms, Software Engineering, Computer Networks, Security Operations, Programming in JavaScript, Advanced Programming in C++, Mobile Application Development, Operating Systems, System Analysis and Design, Project Management.

EXPERIENCE

unWired Broadband **Feb 2023 – Present**
Level 1 Technical Support

- Monitored and analyzed network traffic data to troubleshoot connectivity issues and optimize network performance for over 100+ customers weekly.
- Documented technical procedures and solutions in internal knowledge bases for future reference, improving overall team efficiency and customer satisfaction.
- Developed familiarity with networking hardware, signal troubleshooting, and low-level connectivity diagnostics relevant to IoT systems.

PROJECTS

ESP32 Self-Watering Planter | ESP-IDF | Github
An autonomous planter that is equipped with a submersible pump, light sensor, and soil sensor for sensor based watering.

- Integrated VEML7700 ambient light sensor over I2C — configured bus and device handles, implemented power-on sequence from datasheet.
- Soil moisture sensing via ADC with percentage-based threshold triggering for automated pump control.
- GPIO, MOSFET switching, and inductive load protection configured via ESP-IDF driver APIs.
- Optional WiFi-connected pump controller with browser-based control via HTTP web server hosted on the ESP32 itself.
  Tech: C, ESP-IDF, FreeRTOS, ESP32, PlatformIO, I2C, ADC

STM32 Self-Watering Planter | Baremetal C | Github
An STM32 Based project configured with GPIO, RCC, and MOSFET switching directly via STM32 registers without HAL.

- Implemented pull-down networks and gate resistors for safe inductive load control.
- Learned firsthand why flyback diodes are non-negotiable on inductive loads — the hard way
  Tools used: Baremetal C, ARM Cotrex-M0, STM32CubeIDE, MOSFET Drive

TECHNICAL BLOG - MEDIUM

Documenting embedded systems projects and lessons learned while transitioning into firmware development. Written for beginners entering the embedded space.
Published posts:

- How I Burned My First STM32 (And What I Learned About Flyback Diodes) — MOSFET switching, inductive load protection, hardware debugging.
- How to Connect Simple Digital Modules to an ESP32 — GPIO initialization, signal pin configuration, ESP-IDF.
- How to Connect Analog Modules to an ESP32 — ADC oneshot driver, raw value conversion, sensor calibration.
- I2C Basics with an ESP32, From a Beginner — bus and device handles, register reads, multi-byte data parsing.

SKILLS

Baremetal C, ARM Cortex-M0+, STM32, ESP32, ESP-IDF, FreeRTOS, GPIO, RCC, SPI, I2C, ADC, MOSFET switching, Breadboarding, Soldering, PlatformIO, STM32CubeIDE, VSCode